

NASA/GSFC STATEMENT OF WORK

Extraterrestrial Molecular Indicators of Life Investigation (EMILI) RF Electronics System

February 1, 2025 v.5

NASA CONTACT

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BACKGROUND

The Extraterrestrial Molecular Indicators of Life Investigation (EMILI) project funded by NASA's Maturation of Instruments for Solar System Exploration (MatISSE) program aims to advance the existing EMILI instrument design for potential future planetary missions. EMILI combines gas chromatograph (GC) and capillary electrophoresis (CE) front-end separation subsystems, interfaced to a common linear ion trap mass spectrometer (ITMS) for broad molecular analysis of samples that could reveal signs of extraterrestrial life. The project will develop an Engineering Test Unit (ETU) model of EMILI to be tested in appropriate simulated environmental conditions to achieve technology readiness level (TRL) 6 suitable for accommodation on a remote Ocean World mission.

While the EMILI ITMS is very similar to the Mars Organic Molecule Analyzer (MOMA) and the Dragonfly Mass Spectrometer (DraMS) ion trap instruments, the significant novel element of EMILI is the ion inlet system that provides the interface to the CE subsystem. The ion inlet provides for introduction of electrospray ionization (ESI) particles at high pressure (nominally Earth atmosphere) through a capillary and then sequentially through an ion funnel (IF) and a square quadrupole (SQ) ion optical focusing and transfer assembly. Both IF and SQ require RF voltage control of their electrode assemblies to guide ions through a decreasing pressure gradient and into the ITMS; these control electronics are the subject of this requirement. A more detailed description of the inlet system design and electronics specifications are in the Appendix of this SOW.

REQUIREMENTS

1. Contractor shall support the EMILI project led by GSFC through requirements analysis, design, development, testing, optimization, post-delivery support, and reporting, as appropriate, of the EMILI CE-ESI-MS RF Electronics.

2. Contractor shall design and develop an RF Electronics system, comprising one or more compact RF power supply modules meeting the performance specifications given in the Appendix.
3. Contractor shall seek to minimize RF Electronics size, weight, and power (SWaP) to the degree possible within technical/performance and project scope/duration constraints.
4. The number, size/shape, enclosure, and interface requirements of the RF Electronics module(s) shall be determined jointly by the Contractor and GSFC during a collaborative design requirements/trade study phase. Note: Development and delivery of a flight-like enclosure for the RF Electronics is not required from the Contractor at this time.
5. The RF Electronics shall be compatible with, and meet all performance requirements for, EMILI instrument testing in both laboratory ambient conditions and in planetary T-Vac environmental conditions, comprising high vacuum and a wide temperature range, to be detailed during early-phase collaborative discussions between Contractor and GSFC.
6. Contractor shall provide regular informal status reports, participate in project meetings as appropriate, and provide a complete technical summary at the end of the project to include detailed specifications and designs and all test data generated during the project.

DESIRED FEATURES

7. The RF Electronics should be compatible with planetary mission mechanical environmental loads testing including vibration and shock nominally consistent with General Environmental Verification Standard (GEVS) levels at the EMILI level. Note: Where such testing would depend on incorporation of the RF Electronics in an appropriate flight-like housing mated to/part of the EMILI ETU, the propagated environmental loads on the delivered electronics alone will be taken into consideration.
 - a. It is not required for Contractor to conduct loads testing
8. The RF Electronics should be compatible with dry heat microbial reduction (DHMR) sterilization protocols nominally tolerating 125 °C for at least 10 hours.
 - a. It is not required for Contractor to conduct DHMR processing.

PERIOD OF PERFORMANCE

Start date: March 1, 2025 or as soon as contract award is established

End date: February 28, 2026 (end of baseline EMILI MatISSE PoP); no-cost extensions could be granted depending on programmatic circumstances near the end of the project

NB: Delivery date of hardware from Contractor to GSFC is to be determined during finalization of project schedule following award.

APPENDIX

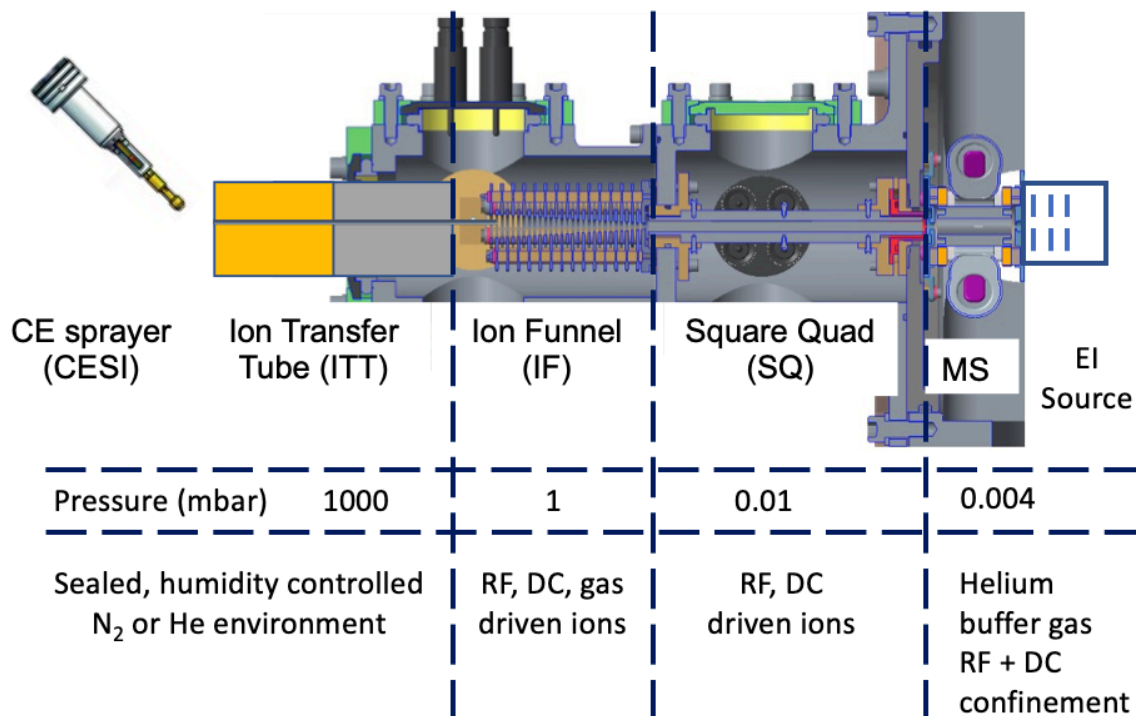
EMILI ESI Source RF Supply Description and Requirements

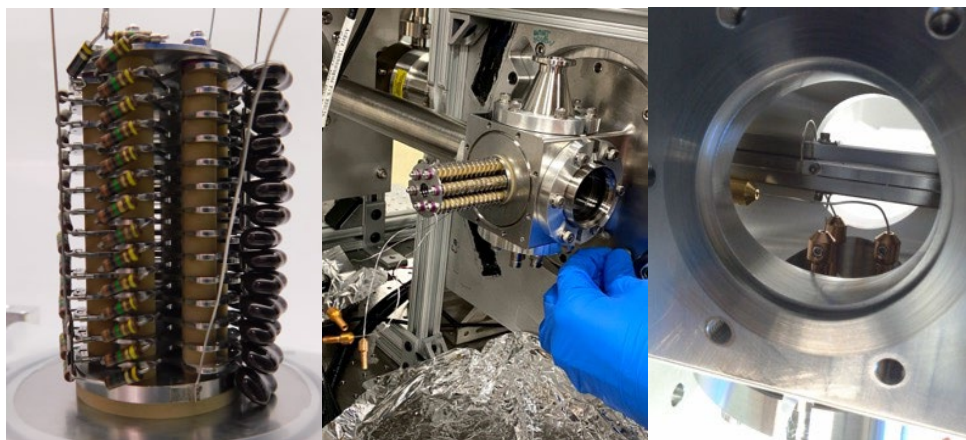
General Description:

EMILI requires an RF supply for two RF based electrospray ionization (ESI) source ion guides. These ion guides are an Ion Funnel and a Square Quadrupole. These devices are “RF only” devices, although they will also have a “lensing” DC bias applied external to the RF supply (via coupling capacitors in the vacuum system).

The Funnel consists of several metal plates and has an internal RC network that should be transparent to the RF circuit. There is a DC bias at the start of the Ion Funnel and at the exit of the Ion Funnel. An RC network provides a DC gradient through the device. Every other plate has an alternating RF phase applied. Plate-to-plate capacitance is $\sim 63\text{pF}$ (in GESI BB MS).

The Square Quad consists of 4 square rods (mounted similarly to a QMS), with the same RF phase applied to opposite rods. The entire rod set has a DC lensing bias/offset applied. Rod-to-rod capacitance is $\sim 18\text{pF}$ (in GESI BB MS).





We are currently considering a trade regarding the definition of the operating frequency for these two RF devices. Instantaneous mass range is our primary science driver that is presumed to be weighed against resource needs. We will either have two different RF frequencies (i.e. two independent RF supplies) or we will run both ion guides at the same RF frequency but at different RF amplitudes (the two amplitudes will be set at a fixed ratio). *An important input to this trade is the overall resources required for a simplified RF supply, so that is hoped to be an early output from this development.*

While this is generically an “RF supply”, the requirements and expected resources for this supply are somewhat different than previous GSFC/PEL/699 RF supplies (i.e. SAM, MOMA, DraMS, etc.). The assumption is that the SWaP of this ESI source RF supply will be significantly smaller than previous analytical RF supplies. We envision this will be a two-way conversation to refine the accommodation within the EMILI ETU.

Per most recent scope discussion discussions it is anticipated that Contractor will explore and likely pursue a design based on Kregel SJ et al. (2021) J. Am. Soc. Mass Spectrom. 32:2821–2826 [<https://doi.org/10.1021/jasms.1c00247>].

Single frequency RF supply requirements:

<u>Function</u>	<u>REQUIREMENT</u>
RF Freq	2 (+/-0.1) MHz
RF Freq Stability	+/-0.1 MHz (TBC)
RF Amplitude Control	30 - 300 Vpp
RF Amplitude Stability	+/-5%
Rod to Rod Balance	+/-5% (TBC)
RF output 1 (Square Quad)	100% RF
RF output 2 (Ion Funnel)	25% RF (TBC)
- Could be accomplished external to the RF supply	

DC Bias (lensing voltage):

Max DC Bias – applied externally	± 200 V
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Two output/frequency RF supply requirements:

<u>Function</u>	<u>REQUIREMENT</u>
RF Freq (Ion Funnel)	2 (+/-0.1) MHz
RF Freq (Square Quad)	3 (+/-0.1) MHz
RF Freq Stability	+/-0.1 MHz (TBC)
RF Amplitude Control - independent	30 - 300 Vpp
RF Amplitude Stability	+/-5%
Rod to Rod Balance	+/-5% (TBC)

DC Bias (lensing voltage):

Max DC Bias – applied externally	± 200 V
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Additional Notes:

- RF frequency is assumed to be self/internally resonant (no external drive option)
- RF frequency *stability* is more important than *accuracy*
- RF Freq/Ampl readback for monitoring (similar to MOMA/LITMS)
- RF supply(ies) need to operate at Earth pressure (lab environment), as well as under vacuum (TVAC and Ocean World)